

CABLE AND PIPE PENETRATION SEALS

CHARCOAT FIRESTOPPING

4 HOUR FIRE RATED ELECTRICAL
CABLE, PVC/STEEL PIPE AND JOINT
PENETRATION SEALS



CHARCOAT

CHARCOAT FM MORTAR



Code: CharCoat FM

Up to 4 Hour Fire and Insulation Ratings

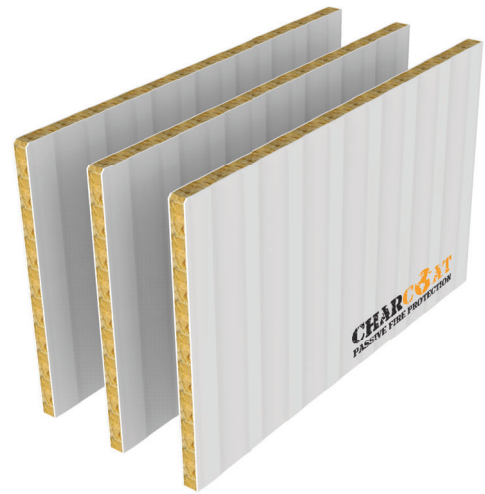
CharCoat FM Mortar is a dry white powder consisting of inorganic compounds and perlite. When mixed with water the compounds form a highly thermal insulating fire seal which prevents the spread of fire and smoke through openings in fire rated walls and floors. This includes openings formed to accommodate building service penetrations.

CharCoat FM expands by up to 1% by hydraulic action during curing, ensuring a very tight seal around service penetrations and the surrounding construction. CharCoat FM has a rapid setting time and it's easy to sand or drill after cure. The compound dries to an off-white colour which may be painted if required.

CharCoat FM will also maintain the acoustic design performance in walls and floors.

CharCoat FM has great strength and adhesion properties with un-paralleled impact testing and the ability to withstand loads of up to 15kN concentrated load and 500Nm serviceability

CHARCOAT FP PANEL



Code: CharCoat FP

Up to 4 Hour Fire and Insulation Ratings

CharCoat FP has been designed to maintain the fire resistance of separating walls/floors where they are breached by single or multiple building services. It consists of a rigid stone wool core, sealed with CharCoat CC Coating on 1 or both faces. The coating provides an important barrier to the passage of fire, smoke & hot gases through the thickness of the board, reducing the rate at which the stone wool core heats up. This provides a better heat sink for the dissipation of conducted heat through penetrations with higher thermal conductivity.

The selection of the board coated on 1 or both faces is determined by installation considerations & fire resistance requirements. On site, CharCoat FP must be used together with CharCoat IS Sealant for sealing around building services and the surrounding construction.

CharCoat FP is a high performance product carrying a classification code up to EI 240 in a wide range of application, making it unique and advanced in a constantly challenging sector. CharCoat FP is also tested in single and double board applications both within solid construction walls, floors and dry lining scenarios. Single board testing has made for faster and more cost effective installation.



CHARCOAT IS SEALANT



Code: CharCoat IS

Up to 4 Hour Fire and Insulation Ratings

CharCoat IS Acrylic Sealant is widely used in markets across Europe, Asia and the Middle East. Its exceptional certification across a wide range of fire-stopping applications makes it popular with installers, reducing the need to carry different mastics. Used in combination with CharCoat FP Panel establishes it as a unique fire rated sealant in the PFP market place.

CharCoat IS has been tested with various types of penetrations through walls, ceilings and floor constructions. Its intumescent property closes any gaps left by burnt away penetrations. It has a shelf life of 18 months.

CharCoat IS is able to cut down application times, minimize material use and cut down on intensive labour costs due to its ability to achieve high fire and insulation ratings with a single sided installation within solid construction elements such as block or concrete walls. It is designed not only for linear seals, but also sealing penetration openings for cables and pipes. It carries classification code up to EI 240 in a wide range of applications.

CharCoat IS is tested and approved for a vast range of services types and installations both within the New Build and Refurbishment Sectors. A few examples are large single cables, large cable bundles and an array of plastic and metal pipes (insulated or un-insulated). Tested in plaster, masonry and concrete constructions and can also be used within dry lining applications. CharCoat IS is one product with the flexibility of multiple applications.

CHARCOAT PS SEALANT



Code: CharCoat PS

Up to 4 Hour Fire and Insulation Ratings

CharCoat PS is a high performance, professional quality, one part, ready to use sealant and adhesive. Based on an innovative new Polymer Technology it is suitable for a wide variety of building trade applications including decorating, flooring, joinery, plumbing and tiling and out-performs conventional silicone, MSP, butyl and acrylic based products as a sealant and adhesive – the only sealant free from dangerous emissions.

CharCoat PS polymer chemistry ensures compatibility with most building materials and has the unique property of moving dynamically to accommodate the natural or unexpected movements of sealed/bonded joints within a building, for long-lasting performance.

CharCoat PS has been tested and approved for many service types and installations performed daily within each phase of construction and within the refurbishment sectors. For instance, joinery, plumbing, flooring, tiling and within many dry lining applications. CharCoat PS can be used as a sealant and an adhesive because its unique chemistry is compatible with most building materials. CharCoat PS, one product for multiple applications.

CharCoat PS uses technology and chemistry to achieve the lowest emission and VOC achievable within the sealants sector, falling well within the requirements of GEV. The results comply with EMICODE emission class EC1PLUS which is the best environmental and indoor hygiene health protection standard achievable.

CHARCOAT GS SEALANT



Code: CharCoat GS

Up to 4 Hour Fire and Insulation Ratings

CharCoat GS is a waterbased, fire classified sealing compound which contains a high concentration of heat expanding graphite material. It also has a shelf life of 18 months.

CharCoat GS is a high specification formulation designed to prevent the spread of fire, smoke and gases through openings in fire rated walls and floors. CharCoat GS expands when it is subjected to heat and closes openings around penetrations when any combustible or low temperature melting materials have burnt away. CharCoat GS is designed to fire seal difficult services which traditional fire rated mastics do not achieve, such as large plastic pipes.

CharCoat GS is a pressure exerting, aperture closer sealant designed to expand on contact with fire/excessive heat. It is resistant not just to the passage of fire, but also the ever more prevalent issue of sound insulation and attenuation. Using CharCoat GS, reductions of up to 53dB RW can be achieved with only a 25mm depth within single sided solid construction substrate, this is higher than most acoustic sealants in the marketplace today

CharCoat GS uses unique technology and chemistry to achieve the lowest emission and VOC achievable within the sealant sector. Falling well within the GEV requirements. The results achieved comply with EMICODE emission class EC1PLUS which is the best environmental and indoor health protection standard achievable.

CHARCOAT ES SEALANT



Code: CharCoat ES

Up to 4 Hour Fire and Insulation Ratings

CharCoat ES Acrylic Sealant is widely used in markets across Europe, Asia and the Middle East for economical fire-stopping applications making it popular with installers for small cable penetrations and vertical / horizontal joint work up to 50mm wide.

CharCoat ES has been tested with various types of penetrations through walls, ceilings and floor constructions for small cables. Its intumescent property closes any gaps left by burnt away penetrations. It has a shelf life of 18 months.

CharCoat ES is able to cut down application times, minimize material use and cut down on intensive labour costs due to its ability to achieve high fire and insulation ratings with a single sided installation within solid construction elements such as block or concrete walls. It is designed not only for linear seals, but also sealing penetration openings for cables and pipes. It carries classification code up to EI 240 in a small range of applications.

CharCoat ES is tested and approved for a small range of services types and installations both within the New Build and Refurbishment Sectors. Ideal for commercial construction.

CharCoat ES has undergone many tests, achieving the markets best European Technical Assessment (ETA) possible, this is necessary to be awarded with the CE mark.

CHARCOAT PIPE WRAP



Code: CharCoat PW

Up to 4 Hour Fire and Insulation Ratings

CharCoat PW Pipe Wrap is a graphite material, within a flexible plastic sleeve that reacts to heat expanding to fill the void created by the melting plastic through penetration. It can be used with plastic pipes (PVC-C, PVC-U, PP, PE, PE-HD, ABS and SAN+PVC), bundles of plastic pipes – with or without cables, and metal pipes with penetrating combustible insulation.

Fitted by wrapping around the through-penetration and securing with the adhesive tape, then casting into the wall or floor construction fitted in either CharCoat FP or CharCoat FM.

CharCoat PW is a flexible sleeve with a graphite intumescent inner, the perfect solution to wrap around plastic and insulated pipes within numerous construction scenarios.

CharCoat PW is very versatile in use and quickly and easily installed. It saves not only time and money, but also the preservation of life and property. Able to cope with a vast array of penetrating elements such as plastic pipes, insulated pipes and cable bundles, this is a light weight, cost effective, but heavy hitting penetration seal.

CharCoat PW has undergone hundreds of tests, achieving the markets best European Technical Assessment (ETA) possible, this is necessary to be awarded with the CE mark.

CHARCOAT CL COLLARS



Code: CharCoat CL

Up to 4 Hour Fire and Insulation Ratings

CharCoat CL Collars consist of a two part, coated, circular steel shell that fits around plastic pipes PVC-U, PVC-C, PE, LDPE, MDPE, HDPE, ABS, SAN+PVC and PP pipes; plastic pipe bundles – with or without cables, and bundles of cables. The collars have a secure and simple locking device. The steel shell contains a patented fast expanding graphite material that reacts to heat. The graphite expands in the event of a fire to fill the opening left by the melting plastic of the through penetration. It can be fitted on the outside of a wall or floor, even in single sided applications.

CharCoat CL is available up to Ø350 mm with a wide range of pipe wall thicknesses. They are also available in two different heights for different fire classifications to maximize cost efficiency. With fire classifications up to 240 minutes for both integrity and insulation.

CharCoat CL is a perfect solution for core drilled pipe and cable bundle penetration holes. The reinforced steel collar exterior is designed to take the pressures of the intumescent's reaction, which expands to fill and crush down the melting pipe. This then fills the void that would otherwise allow for the passage of fire and prevent the passage of unseen toxic gasses.

CharCoat CL fitted as one single side collar on (160mm x 3.2 w/t) PVC pipe, results in a reduction of 58dB airborne sound. Making the CharCoat CL an all-round robust firestop element and sound transfer reduction element.



TRANSPORT STORAGE

- Transport and store free from frost- preferably at a minimum of +5°C to a maximum of +30°C.
- shelf-life of unopened product: 18 month from date of manufacture. Unopened pails must be re-sealed.

PACKAGING

Please refer to the Technical Data Sheet

SURFACE PREPARATION

Please refer to the Technical Data Sheet

APPLICATION

Please refer to the Technical Data Sheet

DRYING TIME

Please refer to the Technical Data Sheet

SAFETY AND ENVIRONMENT

Please refer to the Technical Data Sheet

CHARCOAT

QUICK REFERENCE GUIDE

TYPE OF PENETRATION	SEALS WITHOUT PENETRATIONS						CABLES							STEEL PIPES		COPPER PIPES		ALUPEX PIPES		VENTI-LATION PIPES		PLASTIC PIPES									
	30mm	50mm	100mm	300x300mm	1200x2400mm	1200x∞mm	≤ø21single	≤ø21bundled	≤ø50single	≤ø50bundled	≤ø80single	≤ø80bundled	Trays & Ladders	Without Insulation	≤ø325	Without Insulation	≤ø54	Without Insulation	≤ø75	≤1.2m x 1.7m	≤ø1250	≤ø32	≤ø40	≤ø75	≤ø110	≤ø160	≤ø400	Conduits	PEX pipe-in-pipe		
CHARCOAT ES	✓	✓	x	x	x	x	✓	x	x	x	x	x	x	x	✓	x	✓	x	x	x	x	✓	x	x	x	x	x	x	x	x	
CHARCOAT IS	✓	✓	✓	✓	x	x	✓	✓	✓	✓	✓	✓	x	✓	✓	✓	✓	✓	✓	✓ ₂₎	✓ ₂₎	✓	✓	✓	x	x	x	✓	✓	✓	
CHARCOAT GS	x	x	x	x	x	x	✓	x	x	x	x	x	✓	x	x	x	x	x	x	x	x	✓	✓	✓	✓	✓	x	✓	x	x	
CHARCOAT PS	✓	x	x	x	x	x	x	x	x	x	x	x	x	✓	x	✓	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	
CHARCOAT FM	✓	✓	✓	✓	✓	x	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓ ₂₎	✓ ₂₎	✓	✓	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓	✓ ₁₎	✓
CHARCOAT FP	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓ ₂₎	✓ ₂₎	✓	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓ ₁₎	✓	✓ ₁₎	✓
CHARCOAT PW	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	x	✓	x	✓	x	x	✓	✓	✓	✓	✓	✓	✓	✓	✓	
CHARCOAT CL	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	✓	✓	✓	✓	✓	x	x	x	

✓ = YES | x = NO | ₁₎ with CharCoat PW | ₂₎ with dampers

PRODUCT	FIRE RATING (Hours)
CharCoat FS Mortar	Up to 6 Hours
CharCoat FS Panel	Up to 4 Hours (Single Panel)
CharCoat Intu Seal	Up to 4 Hours

APPROVALS	DESCRIPTION
CharCoat FS Mortar	
BS476, Part 20/22	- Up to 2 hours 75mm
BS476, Part 20/22	- Up to 2 hours 25mm Mortar / 50mm Rockwool / 25mm Mortar
BS476, Part 20/22	- Up to 6 hours 100mm
EN 1366-3	- Up to 4 hour EI240 (insulation) / E240 (integrity)
EN 13501-2	- Up to 4 hour EI240 (insulation) / E240 (integrity)
CharCoat FS Panel	
BS476, Part 20/22	- Up to 4 hours (single Panel 1200 x 600 x 50mm Precoated)
EN 1366-3	- Up to 4 hour EI240 (insulation) / E240 (integrity)
EN 13501-2	- Up to 4 hour EI240 (insulation) / E240 (integrity)
CharCoat IS / PS / GS / ES	
BS476, Part 20/22	- Up to 4 hours Joint seal (wall or floor)
BS476, Part 20/22	- Up to 4 hours Cable/pipe (CharCoat FS Panel backing)
EN 1366-3	- Up to 4 hour EI240 (insulation) / E240 (integrity)
EN 13501-2	- Up to 4 hour EI240 (insulation) / E240 (integrity)
CharCoat CL Collars	
EN 1366-3	- Up to 4 hour EI240 (insulation) / E240 (integrity)
EN 13501-2	- Up to 4 hour EI240 (insulation) / E240 (integrity)
CharCoat PW Pipe Wrap	
EN 1366-3	- Up to 4 hour EI240 (insulation) / E240 (integrity)
EN 13501-2	- Up to 4 hour EI240 (insulation) / E240 (integrity)

NOTE: All Testing and ETA (EN) certification completed UL-eu - Certificates of Conformity



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    **charcoatpfp**

Disclaimer: The above data, particularly the recommendations for the application and use of Charcoat Passive Fire Protection products are based on the manufacturer's knowledge and experience. Due to different materials and conditions of application, which are beyond our control, we recommend in any case to carry out sufficient tests in order to ensure that Charcoat Passive Fire Protection products are suitable for the intended purpose and applications. Therefore, any liability for such recommendations or any oral advice is expressly excluded unless we have acted willfully or by gross negligence. It is always the responsibility of the installer / purchaser to guarantee correct preparation, DFT (Charcoat Coatings) and thickness (Charcoat Firestop Products) of all Charcoat Passive Fire Protection products. Charcoat Passive Fire Protection is not liable for installation or faulty installation. It is always the responsibility of the installer / purchaser to guarantee and certify the installation of materials.

